

Detecting the Undetectable

How Nation-State Attackers Operate Undetected for Years
Inside Critical Infrastructure — and How AI Changes the Outcome

V-Intelligence UEBA: The Behavioral Digital Twin for Cyber Defense

The Threat Landscape Has Changed

Nation-state attackers no longer break in — they blend in. They use legitimate credentials, standard protocols, and built-in system tools. They look exactly like your employees.

287 days

Average time to
detect a breach
(IBM 2024)

\$4.88M

Average cost of
a data breach
(IBM 2024)

5+ years

Salt Typhoon in
US telecom before
discovery

5+ years

Volt Typhoon in
US critical infra
before discovery

The New Reality

These are not hypothetical scenarios. In 2024, the US government disclosed that Chinese state-sponsored groups had been operating undetected inside American telecom and critical infrastructure for years.

Traditional cybersecurity tools — SIEM rules, anomaly detection algorithms, threshold-based alerts — were deployed at every compromised organization. None of them detected the intrusion.

The attackers were not invisible because they were sophisticated. They were invisible because the detection methods are fundamentally blind to their approach.

Case Study: Salt Typhoon — 5 Years Inside US Telecom

What Happened

~2019	Reconnaissance begins against US telecom providers
~2021	Active penetration of telecom infrastructure
2021-2024	Persistent access to AT&T, Verizon, T-Mobile, Lumen, Charter, Windstream
2024	Accessed CALEA lawful intercept / wiretap systems
2024	Harvested call metadata of 1M+ users, including senior US officials
Sept 2024	FBI and CISA discover the breach — not the telecom companies
Oct 2024	Public disclosure; 9 US telecoms confirmed compromised

Why Every Detection System Failed

- ▶ Used legitimate network protocols — no malware signatures to detect
- ▶ Maintained normal traffic volumes — never spiked bandwidth alerts
- ▶ Operated with valid credentials — no brute force, no credential stuffing
- ▶ Exploited lawful intercept systems — outside standard SIEM monitoring
- ▶ CALEA systems managed separately from customer-facing security

✗ Every traditional SIEM rule, IDS signature, and anomaly threshold was in place

✗ None detected the intrusion — for over five years

Case Study: Volt Typhoon — 5 Years in Critical Infrastructure

What Happened

~2019	Initial access to US critical infrastructure networks
2019-2023	Persistent presence in energy, water, telecom, transportation
Ongoing	Pre-positioned for disrupting US-Asia communications
May 2023	Microsoft Threat Intelligence publicly discloses the campaign
May 2023	CISA/NSA/FBI publish Joint Advisory AA23-144A
Feb 2024	CISA confirms 5 years of persistent access in some targets
2024	Follow-up Advisory AA24-038A documents full scope

Why "Living Off the Land" Is Undetectable

- ▶ Zero malware deployed — used only built-in OS tools
- ▶ PowerShell, WMI, cmd, netsh — tools every admin uses daily
- ▶ Authenticated with stolen valid credentials
- ▶ Lateral movement looked identical to IT maintenance
- ▶ No anomalous traffic patterns — everything within normal bounds

✗ EDR, SIEM, IDS all deployed — all failed to detect

✗ Discovered by Microsoft threat researchers, not by deployed tools

Common thread: Both Salt Typhoon and Volt Typhoon were discovered by external researchers — not by any security tool deployed at the compromised organizations. The tools were there. They simply cannot see this type of attack.

The Full Threat Spectrum

Nation-state actors, insiders, and patient adversaries each exploit a different blind spot. No organization faces just one threat category.

Nation-State Telecom

Salt Typhoon

5+ years undetected | Wiretap compromise
9 US telecom providers | Call metadata of senior officials

Nation-State Infrastructure

Volt Typhoon

5+ years undetected | Critical infrastructure
Living Off the Land | Pre-positioned for disruption

Insider Threat

Trusted Employee

8-month gradual escalation | Authorized credentials
All actions individually legitimate | Slow data exfiltration

Slow APT Campaign

Patient Attacker

180-day campaign | Below every threshold
No single anomalous week | Credential harvesting + lateral movement

Each exploits a different blind spot. No single tool covers all four.

Why Every Traditional Approach Fails

Traditional cybersecurity tools are designed for a threat model that no longer exists.

Signature-Based Detection

IDS / Antivirus / EDR

Looks for known malware patterns and attack signatures.

Fails when: Attacker uses no malware — only legitimate tools that are already whitelisted.

Salt Typhoon and Volt Typhoon deployed zero malware. Every file on disk was legitimate.

Rule-Based Detection

SIEM Rules / Correlation

Triggers on predefined conditions — failed logins, privilege escalation, data thresholds.

Fails when: Attacker stays below every threshold. 5% more data each week never triggers a rule.

Every rule at AT&T, Verizon, and 7 other telecoms was in place. None fired for 5+ years.

Statistical Anomaly

Isolation Forest / SVM / LOF

Flags users whose individual metrics deviate from the statistical mean.

Fails when: No single metric is abnormal enough. The attack is in the combination, not any one number.

Our testing: 4 simulated nation-state attacks — Isolation Forest, SVM, and LOF detect zero.

The fundamental problem: These tools ask "is any one number abnormal?" — but modern attackers ensure that no single number ever is. The attack is invisible to every tool that looks at individual metrics.

The Innovation: Behavioral Digital Twin

V-Intelligence UEBA creates a behavioral digital twin for every entity in your environment — a rich, multi-dimensional representation of how each user, device, and application behaves. Not what numbers they generate. How they work.

Traditional: Raw Metrics

- ✗ Monitors individual numbers: login count, bytes transferred, failed auths
- ✗ Each metric checked independently against static thresholds
- ✗ No understanding of relationships between behaviors
- ✗ No context: "50 logins" means the same whether you're IT or Finance
- ✗ Attacker stays below threshold on every metric → invisible

V-Intelligence UEBA: Behavioral Digital Twin

- ✓ Creates a unified behavioral profile capturing all activity dimensions
- ✓ Understands the meaning of actions — not just the magnitude
- ✓ Captures cross-dimensional relationships (access + data + network)
- ✓ Context-aware: knows what "normal" means for each individual role
- ✓ Detects behavioral change patterns invisible to metric-by-metric analysis

The digital twin doesn't ask "is this number abnormal?" — it asks "has this person fundamentally changed how they work?" That is the question Salt Typhoon and Volt Typhoon attackers cannot evade.

Multi-Tier Detection: Progressive Intelligence

V-Intelligence UEBA employs a multi-tier approach — each layer adds detection capability that the previous layer cannot achieve.

Traditional SIEM

Isolation Forest / SVM / LOF

**0 of 4 attacks
detected**

- ▶ Threshold-based anomaly detection
- ▶ Checks individual metrics independently
- ▶ Misses attacks where no single metric spikes
- ▶ Every deployed tool uses this approach

Result: Catches nothing. Every attacker stays below every threshold.

Intermediate Analysis

Statistical Pattern Detection

**1 of 4 attacks
detected**

- ▶ Multi-dimensional statistical analysis
- ▶ Cross-correlates feature combinations
- ▶ Catches louder anomalies that cross statistical bounds
- ▶ Detects Volt Typhoon (noisier LotL campaign)

Result: Catches Volt Typhoon only. 3 stealthier attacks remain invisible.

V-INTELLIGENCE UEBA

Full Behavioral Intelligence

**4 of 4 attacks
detected**

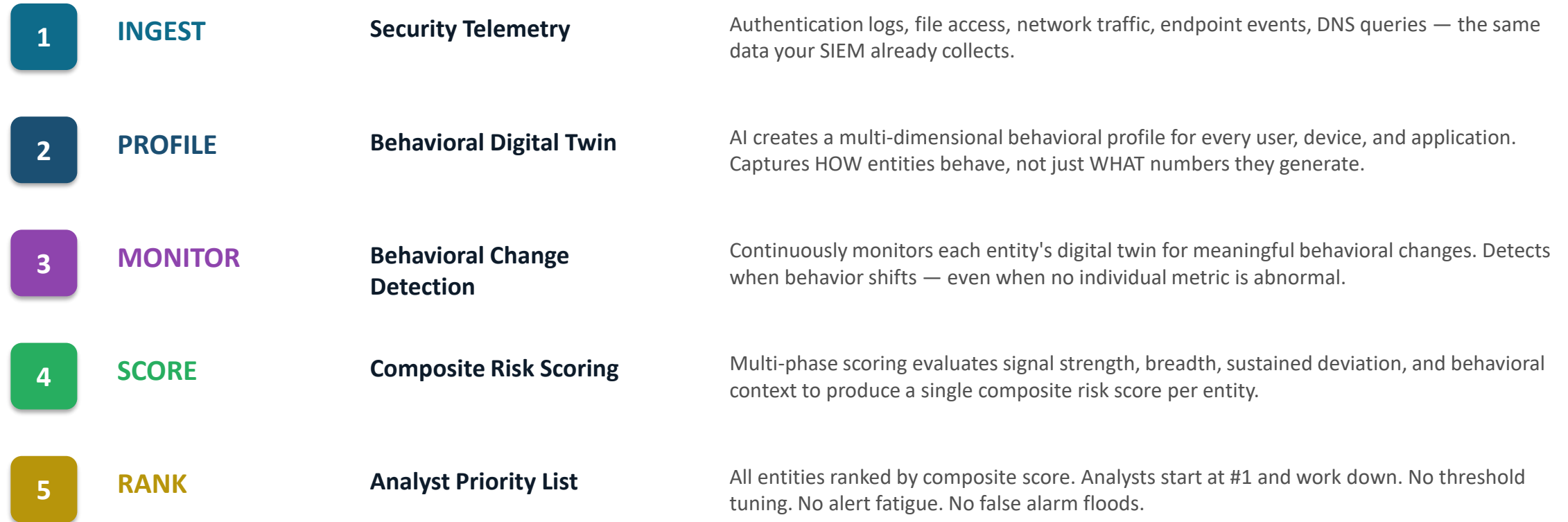
- ▶ Behavioral digital twin per entity
- ▶ Multi-dimensional behavioral profiling
- ▶ Detects subtle pattern shifts across all contexts
- ▶ Catches Salt Typhoon, Insider, Slow APT, Volt Typhoon

Result: All 4 attacks detected at 8.5% false positive rate.

0 / 4 → 1 / 4 → 4 / 4 | Each tier catches what the previous tier cannot see. The behavioral digital twin is the breakthrough layer.

How V-Intelligence UEBA Works — High-Level Architecture

V-Intelligence UEBA transforms raw security telemetry into behavioral intelligence through a multi-stage pipeline:



No new data sources required — V-Intelligence UEBA works with the telemetry your environment already produces.

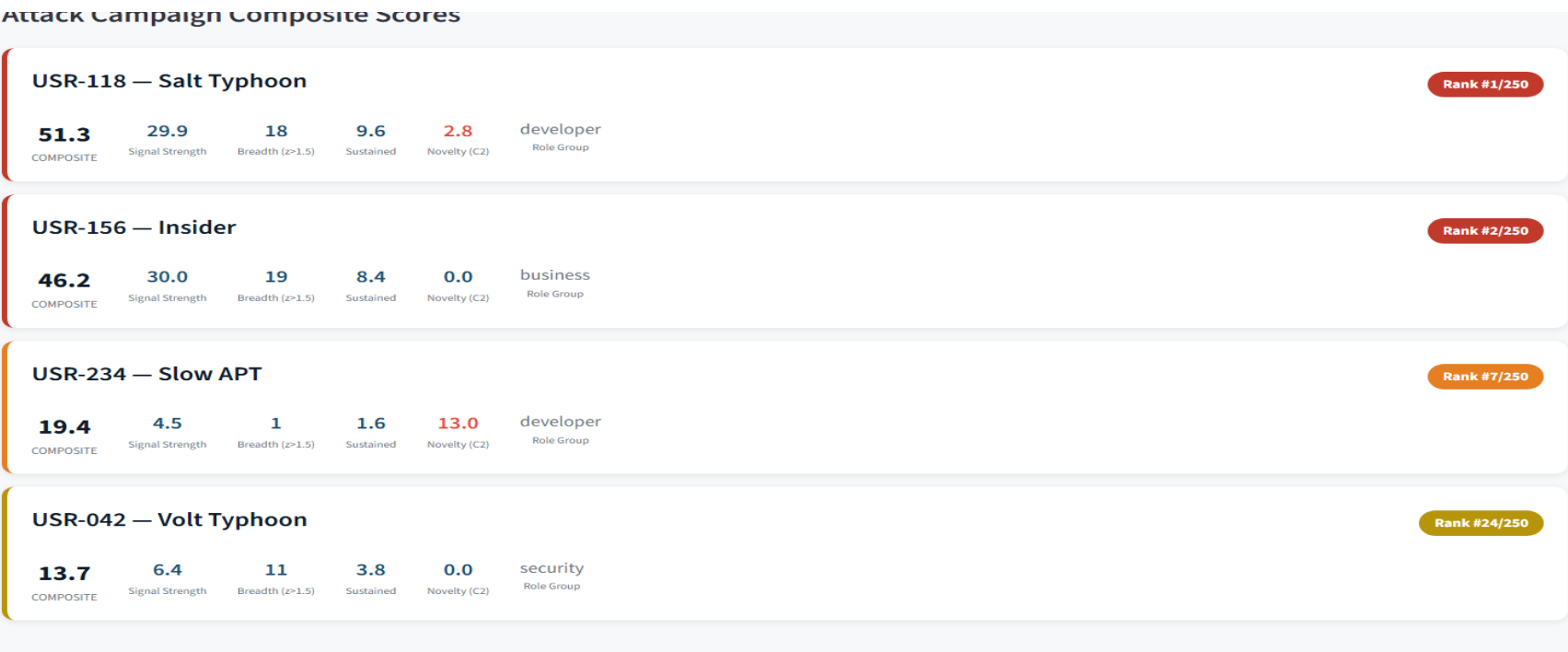
Detection Results: Who Catches What?

250 users, 19 weeks, 4 attack campaigns — traditional methods vs V-Intelligence UEBA:

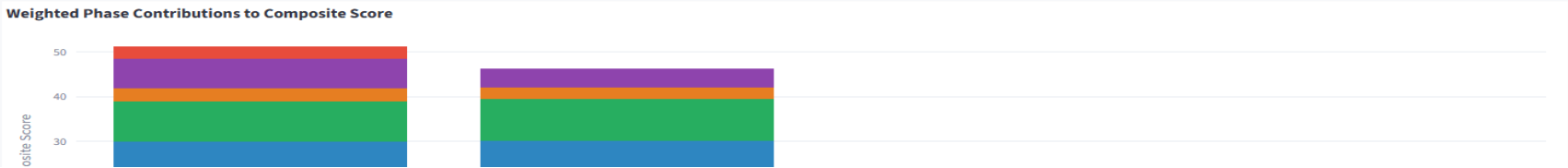
TRADITIONAL DETECTION						0 of 4 detected
	Isolation Forest / One-Class SVM / LOF					
	USR-156 Insider	USR-234 APT	USR-042 Volt Typhoon	USR-118 Salt Typhoon	FP Rate	
Isolation Forest	MISSED	MISSED	MISSED	MISSED	5.3%	
One-Class SVM	MISSED	MISSED	MISSED	MISSED	14.6%	
LOF	MISSED	MISSED	MISSED	MISSED	4.5%	
OUR INTERMEDIATE ANALYSIS						1 of 4 detected
	Statistical Pattern Detection (Z-Score $ z > 3$)					
Z-Score ($ z > 3$)	MISSED	MISSED	DETECTED	MISSED	9.8%	
Catches Volt Typhoon only — Insider, Slow APT, and Salt Typhoon remain invisible						
V-INTELLIGENCE UEBA + COMPOSITE SCORING						4 of 4 detected
	Behavioral Digital Twin → 5-Phase Anomaly Detection					
Composite Score	DETECTED	DETECTED	DETECTED	DETECTED	8.5%	
Traditional: 0 / 4 → Intermediate: 1 / 4 → V-Intelligence UEBA: 4 / 4						
Each tier catches what the previous tier cannot see.						

V-Intelligence UEBA Digital Twin: All 4 Attackers Ranked in Top 10%

Same 250 users, same data. V-Intelligence UEBA's behavioral digital twin identifies every attacker:



5-Phase Composite Breakdown

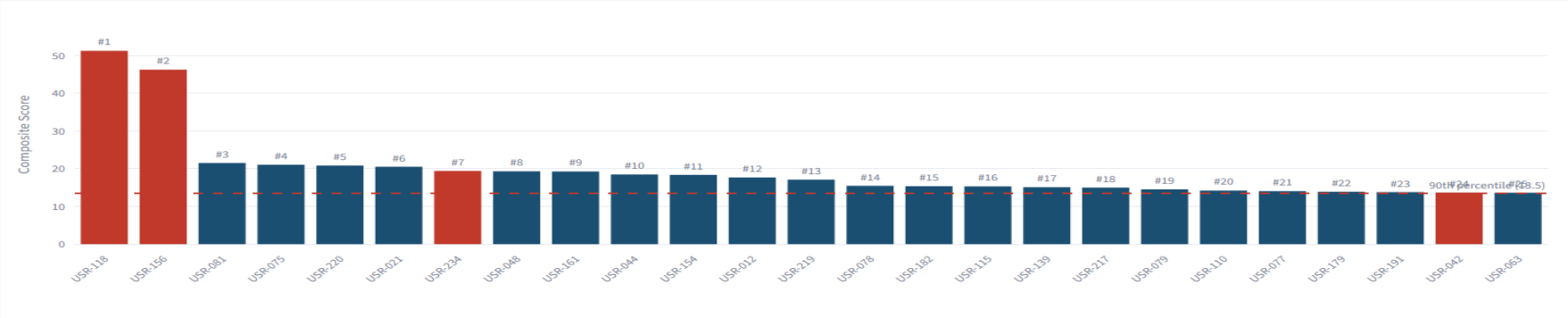


How the Digital Twin Separates Attackers from Normal Users

V-Intelligence UEBA's behavioral analysis creates clear separation that traditional algorithms cannot achieve — attackers are visible in the ranked composite score:

Threshold — Composite score cutoff above which users are flagged. Lowering catches more attackers but increases false positives.
Precision — Of all users flagged, what fraction are actual attackers. Higher is better.
Recall — Of all actual attackers, what fraction were flagged. 100% means no attackers missed.

Top 25 by Composite Score



Novelty Persistence: C2 Beacon Detection



The Verdict: Traditional 0/4 — Intermediate 1/4 — V-Intelligence UEBA 4/4

Same data, same users — three fundamentally different outcomes:

TRADITIONAL DETECTION

Isolation Forest / SVM / LOF

0 of 4

attacks detected

Fixed thresholds on 23 scalar features. Attackers who stay within normal statistical ranges are invisible — no individual metric is abnormal enough to cross a detection boundary.

Best FP: 4.5% (LOF) — but detects nothing

INTERMEDIATE ANALYSIS

Z-Score ($|z| > 3$) — Our Statistical Layer

1 of 4

attacks detected

Detects single-feature spikes beyond 3 standard deviations. Catches the most aggressive attacker (Volt Typhoon) but misses slow, distributed campaigns that stay below the threshold on every individual metric.

FP: 9.8% — catches Volt Typhoon only

V-INTELLIGENCE UEBA

Behavioral Digital Twin + 5-Phase Scoring

4 of 4

attacks detected

Converts telemetry into multi-dimensional behavioral profiles that capture meaning, not just magnitude. Composite scoring fuses five analysis phases into one ranked priority list for analysts.

FP: 8.5% — all 4 attacks detected

V-Intelligence UEBA builds the behavioral digital twin. Composite Scoring detects the anomaly.

4/4 attacks detected at 8.5% FP — vs 0/4 traditional, 1/4 intermediate.

Real-World Validation: Salt Typhoon (USR-118)

5+ years undetected by every traditional tool. V-Intelligence UEBA ranks it #1 out of 250 users (score: 51.3).

Unique Results: What V-Intelligence UEBA Achieved

Four distinct attack campaigns. Four different evasion strategies. All detected and ranked.

USR-118 — Salt Typhoon Telecom Telecom infrastructure pivot — router config exfiltration, call metadata harvesting, DNS tunneling	51.3 COMPOSITE	#1 / 250 100th percentile	Mirrors real Salt Typhoon — 5+ years undetected worldwide
USR-156 — Insider Threat Gradual privilege escalation — restricted file access, off-hours activity, slow data exfiltration	46.2 COMPOSITE	#2 / 250 99th percentile	Slow 8-month escalation — no single week anomalous
USR-234 — Slow APT Low-and-slow persistent access — credential harvesting, lateral movement across network segments	19.4 COMPOSITE	#7 / 250 97th percentile	180-day campaign — deliberately stays below every threshold
USR-042 — Volt Typhoon LOTL Living-off-the-land — PowerShell, WMI, legitimate tools, no malware deployed	13.7 COMPOSITE	#24 / 250 90th percentile	Mirrors real Volt Typhoon — only built-in OS tools

All 4 attackers in top 10% | 8.5% false positive rate | Zero threshold tuning | Traditional methods: 0 of 4 detected

Real-World Validation: Salt Typhoon

Our simulation replicates Salt Typhoon's exact attack pattern. The results confirm reality.

In the Real World

- ▶ Salt Typhoon operated inside US telecom for 5+ years
- ▶ Accessed wiretap systems of senior government officials
- ▶ 9 telecom providers compromised (AT&T, Verizon, T-Mobile...)
- ▶ Discovered by FBI/CISA — not by any deployed tool
- ▶ Every SIEM rule, IDS, and anomaly algorithm was in place

In Our Simulation

- ▶ Same attack pattern simulated among 250 users
- ▶ All traditional algorithms: MISSED (max z-score = 1.71)
- ▶ V-Intelligence UEBA Composite Score: 51.3
- ★ V-Intelligence UEBA Rank: #1 out of 250 users (100th percentile)
- ★ Strongest behavioral anomaly in the entire population

The Bottom Line

The same threshold-based detection that failed for 5+ years in the real world fails identically in our simulation. V-Intelligence UEBA's behavioral digital twin ranks the identical attack pattern as the #1 anomaly — not because any single number is abnormal, but because the digital twin understands that the overall behavior has fundamentally changed.

5+ years

undetected by
traditional tools

#1 / 250

V-Intelligence UEBA rank
(highest anomaly)

51.3

composite
behavioral score

0 of 4

traditional methods
that detected it

Why V-Intelligence UEBA Succeeds Where Others Fail

Every existing approach has a structural blind spot. V-Intelligence UEBA's behavioral digital twin eliminates it.

Traditional SIEM

Checks thresholds on individual metrics

Blind Spot

Misses attacks where no metric exceeds its threshold

V-Intelligence UEBA Advantage

Understands behavior holistically — no threshold to stay below

EDR / XDR

Monitors endpoint activity for known patterns

Blind Spot

Misses credential-based attacks using legitimate tools

V-Intelligence UEBA Advantage

Profiles the entity, not the tool — detects behavioral change regardless of tooling

Network Detection (NDR)

Inspects traffic for signatures and anomalies

Blind Spot

Misses attacks using legitimate protocols at normal volumes

V-Intelligence UEBA Advantage

Captures network behavior as one dimension of a multi-dimensional behavioral twin

Existing UEBA

Statistical deviation on aggregated user metrics

Blind Spot

Fails when no single metric deviates enough — the "combination" problem

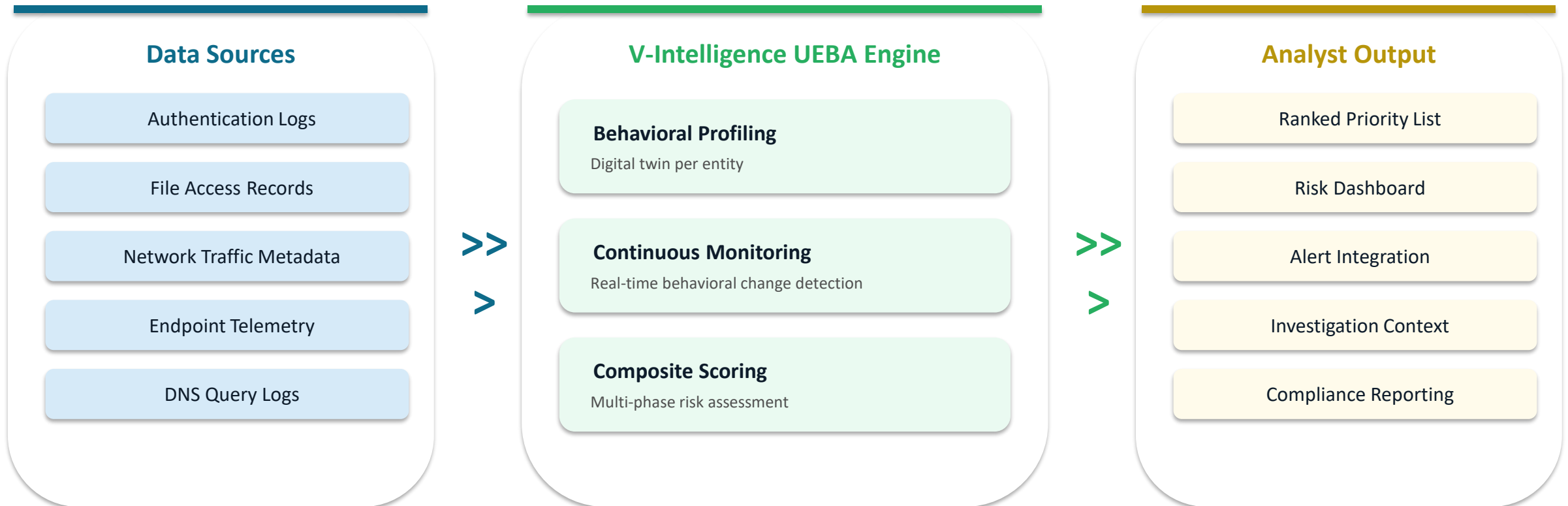
V-Intelligence UEBA Advantage

Digital twin captures cross-dimensional behavioral patterns that scalar metrics miss

V-Intelligence UEBA is not an incremental improvement — it is a new class of detection. The behavioral digital twin sees what no threshold-based system can.

Deployment Architecture

V-Intelligence UEBA integrates with your existing security infrastructure — no new sensors, no new data collection agents.



Cloud or on-premise deployment | Works with existing security infrastructure | No new sensors required

Federal & Critical Infrastructure Alignment

V-Intelligence UEBA directly addresses the mandates and frameworks that govern federal and critical infrastructure cybersecurity.

Executive Order 14028

Improving the Nation's Cybersecurity

Mandates enhanced detection capabilities and continuous monitoring across federal agencies. Requires visibility into advanced persistent threats that evade traditional tools.

NIST SP 800-53

Continuous Monitoring Controls

Behavioral analytics for insider threat and APT detection. V-Intelligence UEBA provides the continuous behavioral monitoring that AU, SI, and CA control families require.

NIST SP 800-207

Zero Trust Architecture

Continuous behavioral verification beyond initial authentication. The behavioral digital twin provides ongoing trust assessment that Zero Trust demands.

CISA LOTL Guidance

Living Off the Land Mitigation

Directly addresses Salt Typhoon and Volt Typhoon TTPs. V-Intelligence UEBA detects the behavioral patterns of credential-based, tool-free attack campaigns.

V-Intelligence UEBA provides the behavioral intelligence layer that existing perimeter and endpoint tools cannot.

Business Impact: What V-Intelligence UEBA Means for Your Organization

Detect Nation-State Threats

Identify Salt Typhoon-class and Volt Typhoon-class attacks that have operated undetected at major US organizations for years. The behavioral digital twin detects the threats that no existing tool can see.

4 of 4 attack types detected

Eliminate the Detection Gap

Close the gap between attacker dwell time (287 days industry average) and detection. V-Intelligence UEBA's digital twin identifies behavioral anomalies within weeks, not years.

Weeks vs 5-year dwell time

Reduce SOC Burden

One ranked list replaces thousands of uncorrelated alerts. Your analysts investigate the highest-risk entities first — not the loudest false alarms.

Single ranked priority list

Future-Proof Detection

The behavioral digital twin adapts as user behavior naturally evolves. No signature updates, no threshold tuning, no rules to maintain.

Zero manual tuning required

V-Intelligence UEBA detects the threats that made headlines — before they make yours.

Salt Typhoon: 5 years undetected.

Volt Typhoon: 5 years undetected.

V-Intelligence UEBA: Detected immediately.

The Behavioral Digital Twin for Cyber Defense
AI-Powered Intelligence for Threats That Traditional Tools Cannot See

4 / 4

attack types
detected

#1 / 250

Salt Typhoon
ranked highest

8.5%

false positive
rate

Zero

threshold tuning
required